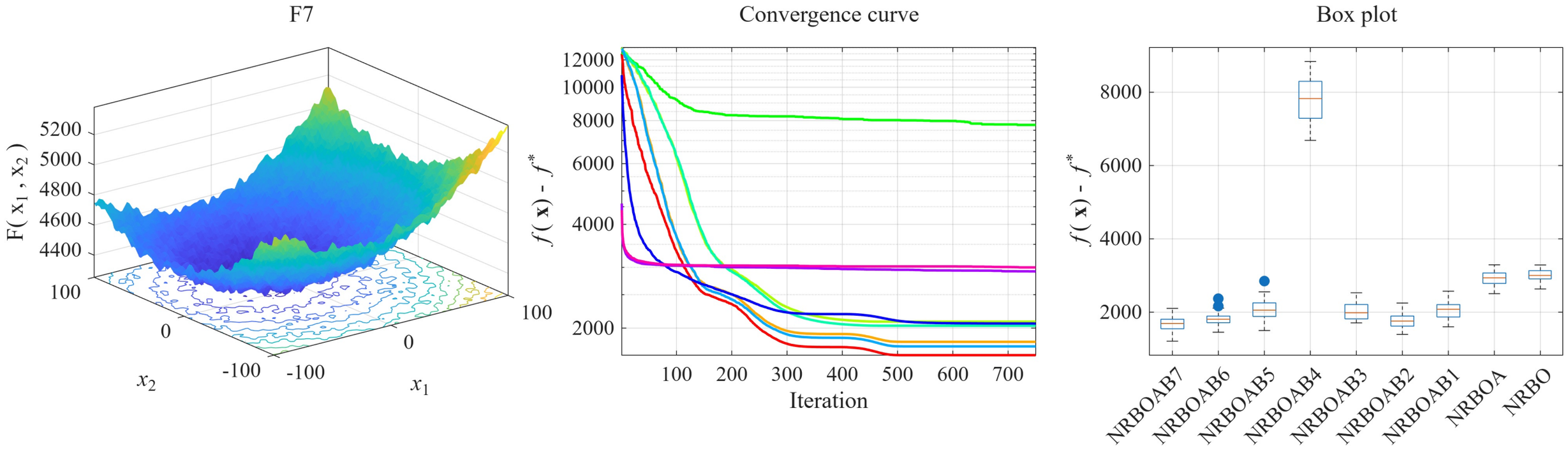

F7
F(x1 , x2)
5200
5000
4800
4600
4400
100
0
-100
x2
-100
0
x1
Convergence curve
f(x) - f*
12000
10000
8000
6000
4000
2000
100
100
200
300
400
500
600
700
Iteration
Box plot
f(x) - f*
8000
6000
4000
2000
NRBOAB7
NRBOAB6
NRBOAB5
NRBOAB4
NRBOAB3
NRBOAB2
NRBOAB1
NRBOA
NRBO
NRBOAB7
NRBOAB6
NRBOAB5
NRBOAB4
NRBOAB3
NRBOAB2
NRBOAB1
NRBOA
NRBO